James A. Michaelov

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Academic Positions

Postdoctoral Researcher: University of Oxford 2026 - Present
Department of Experimental Psychology (Faculty Mentor: **James Whittington**)

Postdoctoral Associate: Massachusetts Institute of Technology 2024 - 2026
Department of Brain and Cognitive Sciences (Faculty Mentor: **Roger Levy**)
MIT Libraries CREOS (Mentor: **Micah Altman**)

Education

Cognitive Science (PhD): University of California San Diego 2024
Thesis: *Understanding the role of statistics in the predictive processing of language*
Committee: **Benjamin Bergen**, Seana Coulson, Marta Kutas, Federico Rossano, Victor Ferreira

Cognitive Science (MSc), with Distinction: University of Edinburgh 2018
Thesis: *Converting from Universal Dependencies to Discourse Representation Theory*
Supervisor: **Federico Fancellu**

Philosophy and Linguistics (MA Hons), First Class Honours: University of Edinburgh 2017
Thesis: *How universal are prominence hierarchies? Evidence from native English speakers*
Supervisors: **Hannah Rohde** & **Jennifer Culbertson**

Publications

Michaelov, J. A., Amo Alonso, C., Chang, T. A., & Levy, R. P. (in press). Target-Language Generation in Multilingual Models: Activation Steering and Optimal Control. *The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*.

Salhan, S., Arnett, C., **Michaelov, J. A.**, & Buttery, P. (in press). Beetle: A Bilingual Model Suite for Modelling Second-Language Processing. *The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*.

Trott, S., **Michaelov, J. A.** Jones, C. R., Chang, T. A., & Bergen, B. K. (2026). Large Language Models as Distributional Baselines for Language Tasks. *Open Mind, 10*, 1213–1252.

Trott, S., Taylor, S. M., Jones, C. R., **Michaelov, J. A.**, & Rivière, P. D. (2026). Language Statistics and False Belief Reasoning: Evidence from 41 Open-Weight LMs. *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)*, 39816—39833.

Michaelov, J. A. & Bergen, B. K. (2026). Better Language Models Better Model the N400, but not Reading Time. *Journal of Memory and Language, 149*, 104762.

Michaelov, J. A., Levy, R. P., & Bergen, B. K. (2025). Language Model Behavioral Phases are Consistent Across Architecture, Training Data, and Scale. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 102407–102447.

Michaelov, J. A., Estacio, R., Zhang, Z., & Bergen, B. K. (2025). Not quite Sherlock Holmes: Language model predictions do not reliably differentiate impossible from improbable events. *Findings of the Association for Computational Linguistics: ACL 2025*, 13528–13551.

Arnett, C., Chang, T. A., **Michaelov, J. A.**, & Bergen, B. K. (2025). On the Acquisition of Shared Grammatical Representations in Bilingual Language Models. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*, 20707–20726.

Michaelov, J. A., Arnett, C., & Bergen, B. K. (2024). Revenge of the Fallen? Recurrent Models Match Transformers at Predicting Human Language Comprehension Metrics. *The Proceedings of the First Conference on Language Modeling (COLM)*.

Michaelov, J. A. & Bergen, B. K. (2024). On the Mathematical Relationship Between Contextual Probability and N400 Amplitude. *Open Mind, 8*, 859–897.

Michaelov, J. A., Bardolph, M. D., Van Petten, C. K., Bergen, B. K., & Coulson, S. (2024). Strong Prediction: Language model surprisal explains multiple N400 effects. *Neurobiology of Language, 5*(1), 107–135.

Michaelov, J. A., Arnett, C., Chang, T. A., & Bergen, B. K. (2023). Structural priming demonstrates abstract grammatical representations in multilingual language models. *The 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023)*, 3703–3720.

Michaelov, J. A. & Bergen, B. K. (2023). Emergent inabilities? Inverse scaling over the course of pretraining. *Findings of the Association for Computational Linguistics: EMNLP 2023*, 14607–14615.

Michaelov, J. A. & Bergen, B. K. (2023). Ignoring the alternatives: The N400 is sensitive to stimulus preactivation alone. *Cortex, 168*, 82–101.

Michaelov, J. A. & Bergen, B. K. (2023). *Rarely* a problem? Language models exhibit inverse scaling in their predictions following *few*-type quantifiers. *Findings of the Association for Computational Linguistics: ACL 2023*, 14162–14174.

Rezaii, N., **Michaelov, J. A.**, Josephy-Hernandez, S., Ren, B., Hochberg, D., Quimby, M., & Dickerson, B. C. (2023). Measuring Sentence Information via Surprisal: Theoretical and Clinical Implications in Nonfluent Aphasia. *Annals of Neurology, 94*, 647–657.

Trott, S., Jones, C. R., Chang, T. A., **Michaelov, J. A.**, & Bergen, B. K. (2023). Do Large Language Models know what humans know? *Cognitive Science, 47*(7), e13309.

Michaelov, J. A. & Bergen, B. K. (2022). Collateral facilitation in humans and language models. *Proceedings of the 26th Conference on Computational Natural Language Learning (CoNLL 2022)*, 13–26.

Michaelov, J. A. & Bergen, B. K. (2022). Do language models make human-like predictions about the coreferents of Italian anaphoric zero pronouns?. *Proceedings of the 28th International Conference on Computational Linguistics (COLING 2022)*, 1–14.

Michaelov, J. A., Coulson, S., & Bergen, B. K. (2022). So Cloze yet so Far: N400 Amplitude is Better Predicted by Distributional Information than Human Predictability Judgements. *IEEE Transactions on Cognitive and Developmental Systems, 15*(3), 1033–1042.

Michaelov, J. A. & Bergen, B. K. (2020). How well does surprisal explain N400 amplitude under different experimental conditions?. *Proceedings of the 24th Conference on Computational Natural Language Learning (CoNLL2020)*, 652–663.

Michaelov, J. A. (2017). The Young and the Old: (T) Release in Elderspeak. *Lifespans and Styles 3*(1), 2–9.

Preprints

Michaelov, J. A., Arnett, C., Chang, T. A., Rivière, P. D., Taylor, S. M., Jones, C. R., Trott, S., Levy, R. P., Bergen, B. K., & Altman, M. (2026). How Open Must Language Models be to Enable Reliable Scientific Inference?. *arXiv:2603.26539*.

Non-Archival Refereed Conference Papers

Michaelov, J. A. & Levy, R. P. (2026). N-gram-like Language Models Predict Naturalistic Reading Time Best. *Proceedings of the Annual Meeting of the Cognitive Science Society, 48*. Rio de Janeiro, Brazil.

Michaelov, J. A. & Arnett, C. (2025). Disaggregation Reveals Hidden Training Dynamics: The Case of Agreement Attraction. *First Workshop on CogInterp: Interpreting Cognition in Deep Learning Models*. San Diego, USA.

Michaelov, J. A., Coulson, S., & Bergen, B. K. (2023). Can Peanuts Fall in Love with Distributional Semantics?. *Proceedings of the Annual Meeting of the Cognitive Science Society, 45*. Sydney, Australia.

Michaelov, J. A. & Bergen, B. K. (2022). The more human-like the language model, the more surprisal is the best predictor of N400 amplitude. *NeurIPS 2022 Workshop on Information-Theoretic Principles in Cognitive Systems (InfoCog)*. New Orleans, USA. (**Spotlight**)

Jones, C. R., Chang, T. A., Coulson, S., **Michaelov, J. A.,** Trott, S., & Bergen, B. (2022). Distrubutional Semantics Still Can't Account for Affordances. In *Proceedings of the Annual Meeting of the Cognitive Science Society, 44*. Toronto, Canada.

Michaelov, J. A., Bardolph, M. D., Coulson, S., & Bergen, B. K. (2021). Different kinds of cognitive plausibility: why are transformers better than RNNs at predicting N400 amplitude?. *Proceedings of the Annual Meeting of the Cognitive Science Society, 43*. Vienna, Austria.

Conference Presentations and Posters

Michaelov, J. A. & Bergen, B. K. (2026). Better Language Models Better Model the N400, but not Reading Time. *The 18th Annual Meeting of the Society for the Neurobiology of Language (SNL 2026)*. Geneva, Switzerland.

Salhan, S., Arnett, C., **Michaelov, J. A.**, & Buttery, P. (2026). Beetle: A Bilingual Model Suite for Modelling Second-Language Processing. *Crosslinguistic Perspectives on Processing and Learning (X-PPL 2026)*. Bayonne, France.

Michaelov, J. A., & Levy, R. P. (2026). 'On the observed scaling effects in language model predictions of reading time'. *The 39th Annual Conference on Human Sentence Processing (HSP 2026)*. Cambridge, USA.

Michaelov, J. A., & Levy, R. P. (2025). 'The effect of orthographic neighborhood density on reading time in 9 languages'. *The 38th Annual Conference on Human Sentence Processing (HSP 2025)*. College Park, USA.

Rezaii, N., **Michaelov, J. A.**, Josephy-Hernandez, S., Ren, B., Quimby, M., Hochberg, D., & Dickerson, B. C. (2023). 'Measuring Sentence Information via Surprisal: Clinical and Theoretical Implications in Nonfluent Aphasia'. *Society for Neuroscience Meeting (SfN 2023)*. Washington, D.C., USA.

Arnett, C., Chang, T. A., **Michaelov, J. A.**, & Bergen, B. K. (2023). Crosslingual Structural Priming and the Pre-Training Dynamics of Bilingual Language Models. *The 3rd Multilingual Representation Learning Workshop (MRL 2023)*. Singapore.

Michaelov, J. A., Coulson, S., & Bergen, B. K. (2022). 'Do we need situation models? Distributional semantics can explain how peanuts fall in love'. *The 35th Annual Conference on Human Sentence Processing (HSP 2022)*. Santa Cruz, USA.

Michaelov, J. A., Coulson, S., & Bergen, B. K. (2022). 'Cloze behind: Language model surprisal predicts N400 amplitude better than cloze'. *The 35th Annual Conference on Human Sentence Processing (HSP 2022)*. Santa Cruz, USA.

Michaelov, J. A., Bardolph, M. D., Coulson, S., & Bergen, B. K. (2021). 'Is the relationship between word probability and processing difficulty linear or logarithmic?'. *The 34th CUNY Conference on Human Sentence Processing (CUNY 2021)*. Philadelphia, USA.

Michaelov, J. A., Bardolph, M. D., Coulson, S., & Bergen, B. K. (2020). 'Surprisal is a good predictor of the N400 effect, but not for semantic relations'. *AMLaP 2020: 26th Architectures and Mechanisms for Language Processing conference*. Presentation in *Special Session: Computational models of language processing*. Potsdam, Germany.

Michaelov, J. A., Culbertson, J., & Rohde, H. (2017). 'How universal are prominence hierarchies? Evidence from native English speakers'. *AMLaP 2017: 23rd Architectures and Mechanisms for Language Processing conference*. Lancaster, UK.

Michaelov, J. A. (2017). 'The Young and the Old: (t) Release in Elderspeak'. *ULAB 2017: 7th Undergraduate Linguistics Association of Britain Conference*, University of Cambridge, Cambridge, U.K.

Media Interviews

Sandrine Ceurstemont (2023). Bigger, Not Necessarily Better. *Communications of the ACM*. **(Interviewed and quoted in article)**.

Awards, Grants, and Fellowships

Honorable Mention, School of Social Science Dissertation of the Year, UCSD 2025

Cognitive Science Diversity Teaching Assistant Award: for promoting accessibility, diversity of perspective, and an inclusive culture as a Teaching Assistant in the Department of Cognitive Science at the University of California San Diego. 2023

Cognitive Science Society Student Travel Grant ($1,200) 2023

CARTA Annette Merle-Smith Fellowship ($20,000) 2021-2022

Center for Academic Research and Training in Anthropogeny Fellowship ($20,000) 2020-2021

Glushko Travel and Research Account ($500/year) 2018-2022

Invited Talks and Panels

What do language models learn when they learn language?
- *Language and Intelligence Lab*, Massachusetts Institute of Technology. 2026
- *Natural Language Processing Seminar*, Cornell Tech, Cornell University. 2026

What can psycholinguistics and natural language processing learn from each other?
- *Language & Cognition Research Seminar*, Harvard University. 2025
- *Computation and Psycholinguistics Lab*, New York University. 2025
- *Cog Lunch*, Massachusetts Institute of Technology. 2025

Panelist for 'Decoding the Mind: What is thought?'
- *Ideas on Consciousness*, University of California, Berkeley. 2025

What can language models tell us about the N400?
- *Center for Research in Language*, University of California San Diego. 2024
- *Distinguished Speakers in Language Science*, Saarland University. 2024
- *Psycholinguistics Group*, Saarland University. 2024

Using language models to understand the N400 (and vice-versa)
- *Computational Psycholinguistics Laboratory*, Massachusetts Institute of Technology. 2023
- *Language Processing Group*, University of California, Irvine. 2023

Can Peanuts Fall in Love with Distributional Semantics?
Lupyan Lab, University of Wisconsin-Madison. 2023
Cognition at the Shore, University of California San Diego. 2023

Panelist for 'Academic Panel: Postgraduate Studies'
The 8th Undergraduate Linguistics Association of Britain Conference,
The University of Edinburgh. 2018

Student Mentorship

University of California San Diego

Reeka Estacio: *Undergraduate Research Assistant Mentee, Co-author* 2023-2025
Zhien Zhang: *Undergraduate Research Assistant Mentee, Co-author* 2023-2025
Van Nguyen: *Honors Program Graduate Student Advisee* 2024
Norah Kerendian: *Undergraduate Research Assistant Mentee* 2023
Srija Sankavaram: *Undergraduate Research Assistant Mentee* 2023

Teaching Experience

Teaching Assistant: University of California San Diego

Introduction to Python 2024 (Spring Quarter)
Data Science in Practice 2024 (Winter Quarter)
Introduction to Data Science 2023 (Fall Quarter)
Learning, Memory, and Attention 2023 (Spring Quarter)
Neurobiology of Cognition 2023 (Winter Quarter)
Cognitive Consequences of Technology 2022 (Fall Quarter)
Cognitive Perspectives 2022 (Summer Session 2)
*What the *#!?: An Uncensored Introduction to Language* 2021 (Fall Quarter)
Cognitive Neuroeconomics 2020 (Fall Quarter)
Cognitive Neuroeconomics 2020 (Summer Session 2)
Language Comprehension 2020 (Summer Session 1)
Cognitive Neuroeconomics 2020 (Winter Quarter)
*What the *#!?: An Uncensored Introduction to Language* 2019 (Fall Quarter)
Minds and Brains 2019 (Spring Quarter)

Tutor (Teaching Assistant): University of Edinburgh

Logic 1 2018 (Semester 2)
Informatics 1: Computation and Logic 2017 (Semester 1)

Service Roles

Program Committee: Cognitive Modeling and Computational Linguistics Workshop 2023–2025
Graduate Student Representative: Department of Cognitive Science, UCSD 2019–2020
Cognitive Science Department Representative: UCSD Graduate Student Association 2018–2019
National Committee: The Undergraduate Linguistics Association of Britain 2017–2018
Webmaster: The University of Edinburgh Linguistics and English Language Society 2016–2017
Webmaster: The Undergraduate Linguistics Association of Britain 2015–2016
Treasurer: The University of Edinburgh Linguistics and English Language Society 2015–2016

Peer Review Experience

Journals:
Journal of Memory and Language: Reviewer
Language, Cognition and Neuroscience: Reviewer
Northern European Journal of Language Technology: Reviewer
Lifespans and Styles: Reviewer

Conferences:
The Annual Meeting of the Association for Computational Linguistics (ACL): Reviewer, Area Chair
The Annual Meeting of the Cognitive Science Society (CogSci): Reviewer, Meta-Reviewer
The Conference on Computational Natural Language Learning (CoNLL): Area Chair
The Annual Conference on Neural Information Processing Systems (NeurIPS): Reviewer
The Conference on Language Modeling (COLM): Reviewer
The Workshop on Cognitive Modeling and Computational Linguistics (CMCL): Reviewer
The Workshop on Interpreting Cognition in Deep Learning Models (CogInterp): Reviewer

Other Professional Experience

Applied Scientist Intern: Amazon (Alexa Games), *Seattle, WA* 2023
Project: Utilizing and developing state-of-the-art natural language understanding systems.
Skills: Applied research, production-level code, product development, product evaluation.

Tools and Resources Developed

Parallel architecture (Parc) Language Models:
`https://huggingface.co/collections/jmichaelov/parc-models-68111455eee2f8461b8b593e`
Language models of different architectures trained in parallel to facilitate analyses of differences between model architectures and their training dynamics. Currently includes Pythia, Mamba-1, and RWKV-4 models trained on the same dataset with the same 6 random seeds for data shuffling and model initialization, with intermediate training checkpoints provided.

PsychFormers:
`https://github.com/jmichaelov/PsychFormers`
Command-line tool that allows the user to use transformer neural network language models to calculate metrics that are relevant to psycholinguistic experiments. Uses the *transformers* Python package and the OpenAI GPT-3 API.

Easy-FAS:
`https://github.com/jmichaelov/easy-fas`
Command-line tool that allows a user to easily calculate the Forward Association Strength between any two words based on norms gathered from human participants. Uses the *Edinburgh Associative Thesaurus* and the *University of South Florida Free Association Norms*.

Word Embedding Similarity Calculator:
`https://github.com/jmichaelov/word-embedding-similarity`
Command-line tool that allows the user to calculate the similarity between two words, or the similarity between a word and its context, based on word embeddings. Compatible with *GloVe* and *fastText*.

Contributions to Large Open Source Projects

Language Model Evaluation Harness:
`https://github.com/EleutherAI/lm-evaluation-harness`
Framework for the standardized testing of generative language models on a large number of different evaluation tasks. Widely used in research and as the backend for benchmark leaderboards.
Regular Contributor (see `https://zenodo.org/records/18636344`)